

PHSP helicity-angle response through v3 BDT

100k generated events; frozen 1,028-chunk score scenario

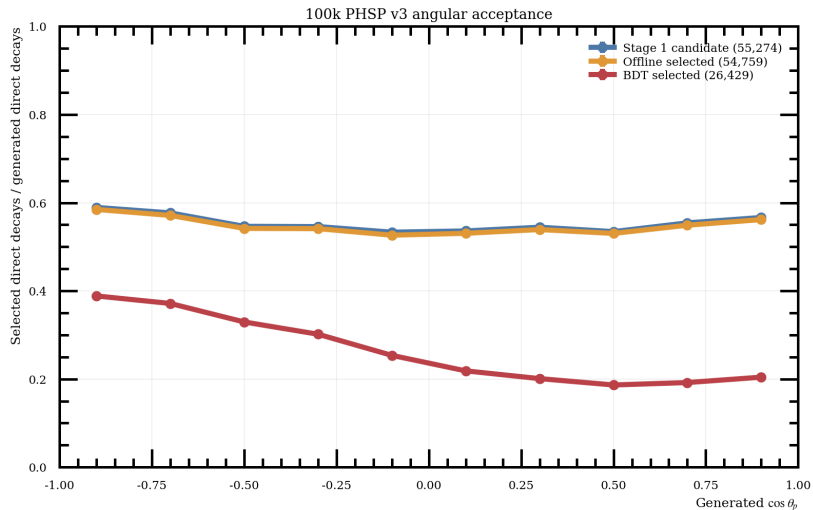
FCC-ee $\Lambda_b \rightarrow \Lambda \gamma$ analysis

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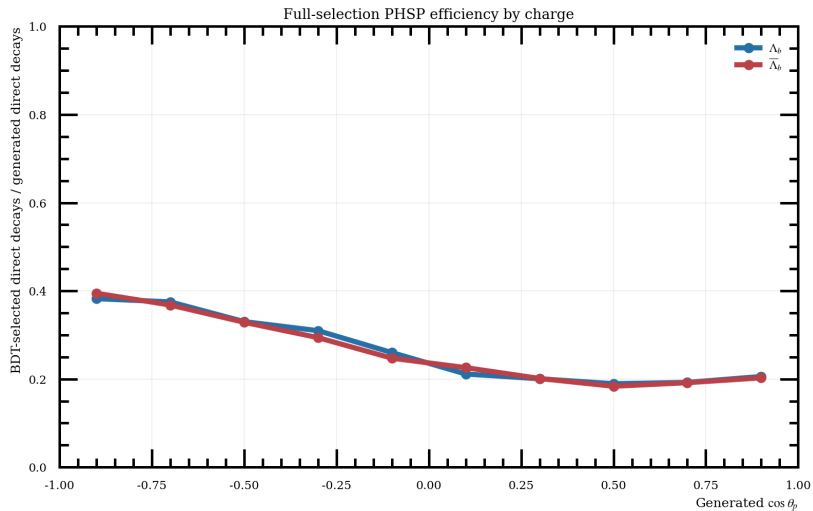
Question, samples and denominators

- Forced PHSP Λ_b and anti- Λ_b : 100,000 input events, 100,007 direct generated decays.
- Existing Stage 1 v3 tuple: 56,737 candidate-bearing output events; 60,426 candidate rows.
- Offline: Λ mass ± 12.5 MeV and $E_{\Lambda_b} \geq 10.5$ GeV, no photon veto.
- Frozen BDT score ≥ 0.9787055 from the 1,028-chunk Physics/Zbb study; PHSP was not in training.
- Efficiency numerator: unique fully truth-matched direct decays in a bin of generated $\cos \theta_p$; denominator: all generated direct decays in that bin.

Cumulative efficiency versus generated angle



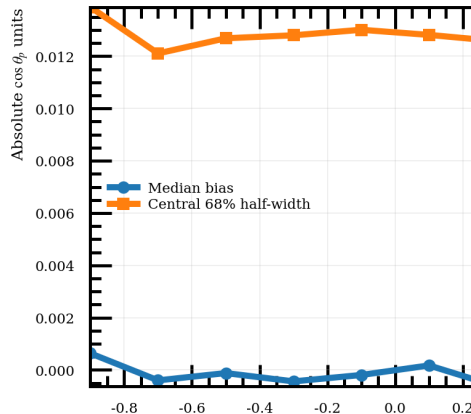
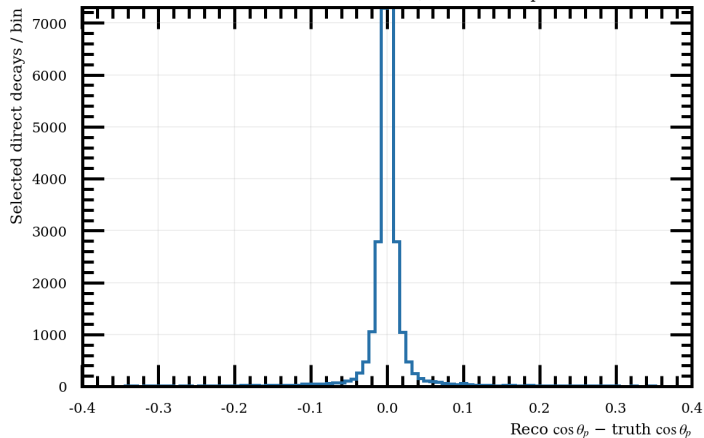
Charge conjugates



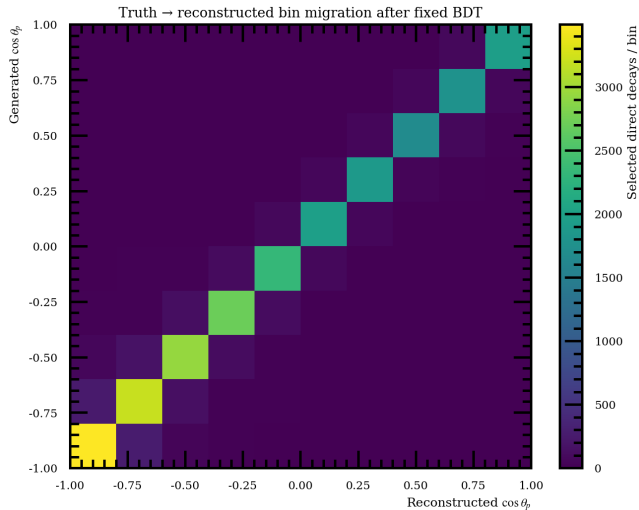
Reconstructed minus generated angle

After offline selection and fixed BDT score

Core shown within ± 0.4 ; see full-tail plot



Migration for a binned angular fit



Numerical summary and fit contract

Stage	Direct candidate rows	Unique direct decays	Cumulative efficiency
Stage 1	55,294	55,274	55.27%
Offline	54,779	54,759	54.76%
Fixed BDT	26,436	26,429	26.43%

- BDT acceptance changes from 0.389 in the most negative truth bin to 0.186 in $0.4 \leq \cos \theta_p < 0.6$.
- Selected residual: median 1.0×10^{-5} , central 68% half-width 0.0118, RMS 0.0888; 5.6% have $|\Delta \cos \theta_p| > 0.1$.
- Fit the generated angular shape after folding through the truth-row/reconstructed-column response matrix. Recompute if the score or model changes.